

# Drone-Augmented Ground Sensors: A Model-Based Systems Engineering Approach to Area Denial

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**Abstract**— Conventional minefields, while effective, are costly, inflexible, and inherently indiscriminate, creating lasting humanitarian and legal challenges under international circumstances. Modern battlefields demand area-denial systems that are precise, adaptive, and ethically compliant. This project applies a Model-Based Systems Engineering (MBSE) approach to design a non-lethal, sensor-based alternative to traditional minefields capable of detecting, classifying, and reporting adversary movement without the use of explosives. Stakeholder analysis incorporating operational, policy, and humanitarian perspectives guided the development of system requirements and logical architecture. An Analysis of Alternatives (AoA) evaluated multiple configurations against criteria including effectiveness, scalability, cost, and disposability. The selected concept integrates a low-power ground sensor network with autonomous drone overwatch to provide adaptive, scalable, and ethically responsible area denial. This paper presents the system's development process, trade-off analysis, and conceptual design, demonstrating a viable framework for future non-lethal area denial capabilities.

**Keywords**— area denial, MBSE, non-lethal systems, autonomous drones

## I. INTRODUCTION

Modern battlefields demand precision, adaptability, and ethical accountability. While traditional area-denial systems, which center around the use of landmines, are fairly effective, they remain costly, rigidly deployable, and inherently indiscriminate. They leave long-term hazards that endanger civilians and complicate post-conflict recovery. To remain effective and responsible in future conflicts, militaries must transition from static explosive obstacles to adaptive, intelligent systems capable of denying terrain without causing unintended harm.

Recent advances in low-power electronics, multi-modal sensing, and autonomous systems enable a fundamental reimagining of area denial. A sensor-based approach can deliver persistent monitoring without explosive hazards by integrating distributed ground sensors with aerial overwatch platforms. These systems can detect and classify intrusions in real time,

coordinate with precision fires when necessary, and safely deactivate after mission completion. By replacing static explosive barriers with intelligent, adaptive networks, modern area denial can achieve tactical effectiveness while maintaining compliance with international humanitarian law. This project leverages model-based systems engineering (MBSE) to systematically design such a system, ensuring traceability from stakeholder needs through conceptual design and prototyping.

This project aims to design and prototype a safe, sensor-based area denial system that leverages distributed multi-modal sensors to track adversary movement and integrates drone overwatch for verification and targeting support. The design process follows an MBSE framework to ensure traceability from mission needs through system requirements and architectural decisions. A comprehensive stakeholder analysis incorporating operational, policy, and humanitarian perspectives guided the development of system requirements and logical architecture. An AoA then evaluated multiple configurations against criteria including effectiveness, scalability, cost, and disposability, leading to the selection of a hybrid ground sensor and drone overwatch concept. This paper presents the system's development methodology, stakeholder analysis, AoA trade-off analysis, conceptual design, initial prototyping efforts, and recommendations for future work.

## II. BACKGROUND

### A. Shaping Terrain Through Area Denial

Shaping terrain through area denial is a fundamental aspect of modern land warfare. By deliberately controlling how and where an adversary can move, commanders can protect key assets, channel enemy forces into engagement zones, and create conditions for decisive fires. Area denial operations centers on the construction of an obstacle belt, most often composed of minefields reinforced with natural or man-made barriers such as wire entanglements or anti-vehicle ditches. These obstacles are carefully sited along likely enemy avenues of approach and integrated into the overall defensive plan. Their purpose is not

only to block or slow movement but also to fix the enemy in place, forcing them to react under unfavorable conditions.

As shown in Figure 1, an effective obstacle belt is always supported by overwatch positions that provide observation and fire control over the denied area [1]. From these vantage points, soldiers monitor the minefield, detect approaching forces, and communicate target data to indirect fire assets. Once an enemy element begins breaching operations, the overwatch unit can radio for artillery, mortar, or air-delivered fires to strike the exposed force while it is slowed or halted by the obstacle. This coordination between obstacle and overwatch turns the minefield from a passive barrier into an active component of the defense, ensuring that any attempt to penetrate it results in concentrated, accurate firepower.

Breaching an area denial obstacle is a deliberate and dangerous process. Attacking forces must first identify the minefield, mark its boundaries, and employ specialized engineers to clear lanes under fire. Breach operations often involve mechanical plows, explosive line charges, and mine rollers, each vulnerable to enemy observation and artillery. The goal of the defender is to disrupt the breach to break the attacker's momentum or destroy key assets while they are confined to narrow cleared paths. Area denial therefore serves multiple purposes: it delays enemy advance, channels movement into predictable routes, protects friendly forces, and enables precise engagement with supporting fires.



Fig. 1. Schematic of the current area denial process.

### B. Importance of Area Denial Systems on the Modern Battlefield

Area denial has long been a central element of land-based warfare for state forces around the world. At its core, area denial is intended to “divert, disrupt, delay, or damage” an enemy to allow friendly forces to gain advantage in maneuver and time [2]. Traditionally, minefields are the most widely recognized form of area denial, used in a vast variety of ways. It is necessary to refer to recent cases to describe the effects of minefields as effective and versatile area denial systems.

Likely, the singular most notable demonstration of landmine usage as an area denial system can be found in Korea. Landmines have been in Korea, particularly the Korean Demilitarized Zone (DMZ), ever since the Korean War ended in 1953. The DMZ today is still heavily influenced by landmines,

and mainly utilized to help separate the Korean borders [3]. In over 70 years, the DMZ has only been breached a handful of times, and its security is largely in part to the minefield, which slows down intruders enough to accurately and efficiently deal with them using lethal force.

Another of the most contemporary and relevant examples is the current Russian Ukrainian conflict. Russia has deployed at least 13 types of antipersonnel mines since February 24, 2022, in at least 11 different regions of Ukraine. The Russians have viewed their landmine usage “as an effective way of ensuring the security of Russia’s borders” [4]. They have additionally made a habit of retreating while emplacing ordinance in strategic positions; this enables booby traps, which is creating complications for the Ukrainian counter-offensive. In general, however, landmine use can prevent not just the military from traversing areas, but civilians from accessing homes, infrastructure, transportation routes, and agricultural lands. In total, 174,000 square kilometers of Ukrainian land is now contaminated with landmines, which has severely limited how ground forces can travel [5]. As a tool in land-based warfare, used to either divert, disrupt, delay, or damage the enemy, landmines in the Russian Ukrainian conflict are performing their mission perfectly, and have shaped the course of the conflict more effectively than even drones.

### C. Current Systems

Current area denial systems rely on minefields that incorporate both anti-personnel and anti-tank mines that achieve the Army tactical mission tasks of blocking, fixing, turning, and disrupting enemy maneuver. Anti-tank mines use large explosive charges that can disable or destroy vehicles, preventing them from moving through an area. Common anti-tank mines are the American M15 and M19 or the Russian TM-62 and TM-83. Anti-personnel mines use smaller explosive charges meant to incapacitate infantry. Common anti-personnel mines include the American M14 and M16 bounding mines and the Russian PMN-2 and MON-50. When used in tandem, anti-tank mines prevent vehicles from moving through an area, forcing dismounted troops to clear a safe lane which is prevented by the anti-personnel mines. It is important to note that there are no limits to the use of anti-tank mines. Meanwhile, countries that did not sign the Ottawa Treaty, which includes the U.S. and Russia, are still allowed to use anti-personnel mines.

These mines can be hand emplaced by engineers or deployed quickly across larger areas using mechanical or artillery-based scatterable systems, commonly referred to as scatterable mines (SCATMINES). The United States deploys SCATMINES with the M87 Rapid Anti-Armor Mine System (RAAMS) and the M718 and M741 Area Denial Artillery Munitions (ADAMS) [6]. Russia uses comparable systems with the ISDM Zemledelie and the Grad and Uragan multiple rocket launchers equipped with remote mining warheads [7].

The minefield is only one component of the overall area denial system. The system also includes observers who are positioned to overwatch the minefield and detect enemy movement. They maintain radio communication with artillery and maneuver units, so they can call for artillery fire on enemy formations as they attempt to breach or bypass the obstacle. The artillery unit is therefore also a key part of the area denial system.

It is important to note that in armed conflict, mines almost never kill enemy combatants. The artillery does.

### *Issues With Current Systems*

For a traditional minefield-based system to be used both properly and efficiently, an element must conduct overwatch duty. The overwatch provides real-time information to the commander, which in turn enables timely and informed battlefield decisions. However, the overwatch position is highly vulnerable to enemy attacks, especially artillery and, in today's era, drone strikes. This vulnerability represents a significant weakness in the current minefield employment system.

Another issue involves the speed of response of traditional minefields. They are not quick to deploy, nor are they easy to rearrange or repurpose once emplaced. Speed of response is a critical aspect of military operations and can mean the difference between mission success and failure, as well as between safeguarding or endangering military personnel. As it stands, current minefields are not versatile enough to respond quickly to the tempo demanded on the modern battlefield, which puts forces at risk.

The difficulty of repurposing minefields also contributes to the challenge of completely demilitarizing them so that they do not become a humanitarian hazard. Mines today, such as SCATMINES, attempt to mitigate this by incorporating self-destruct features. They are designed to last between 4 hours and 15 days before self-destructing. This approach helps address the humanitarian concerns associated with minefield deployment, although it is imperfect due to the risk of dud munitions, and it reduces minefield longevity.

Conventional minefields that do not self-destruct pose significant long-term disposal challenges. Not all minefields are marked correctly and, as a result, can remain active for decades. According to the Monitoring and Research Committee, a global network of NGOs, researchers, and field experts coordinated under the International Campaign to Ban Landmines (ICBL), "At least 5,757 casualties of landmines and explosive remnants of war (ERW) were recorded (1,983 killed and 3,663 injured) for 2023. The survival status was unknown for 111 casualties. In 2023, mine and ERW casualties were identified in 53 states and two other areas. Of these, 38 are States Parties to the Mine Ban Treaty. Civilians made up 84 percent (4,335) of all recorded casualties where the military or civilian status was known. Children accounted for 37 percent (1,498) of civilian casualties where the age group was recorded" [8].

While mines offer a convenient and effective tool for achieving area denial and supporting specific tactical objectives, their deployment creates severe long-term disposal challenges that can continue to harm innocent people long after the fighting has ended.

### III. METHODOLOGY

This project uses an MBSE approach to guide the development of a safe, sensor-based area denial alternative that improves tactical effectiveness while removing humanitarian risk. This system is termed as the Future Area Denial System. MBSE is the formalized application of modeling to support

system requirements, design analysis, verification, and validation activities beginning in the conceptual design phase and continuing throughout development and later life cycle phases. MBSE is a practice in which the primary artifact of systems engineering is an integrated, coherent, and consistent system model rather than a disconnected set of documents and diagrams [9]. MBSE provides the structure needed to capture stakeholder requirements, define system architectures, and evaluate design trade-offs throughout development.

Figure 2 displays a flowchart of the methodology used for this study. The backbone of this analysis is a system model including a requirements analysis, logical architecture, and a physical architecture. These models are developed in Dassault Systems of Systems Architecture using Systems Modeling Language (SysML).

The study started with a comprehensive stakeholder analysis to identify the needs of the system. It incorporated operational, policy, and humanitarian perspectives to guide requirement definition and ensure the system aligns with both mission objectives and ethical standards. The stakeholder needs provide inputs to the model in the form of operational requirements, which indicate what the system must be able to do in an operational context. The requirements analysis provided a series of metrics that can be used to evaluate the performance of the system. It also provided a set of high-level functions that the system must perform to meet the stakeholder needs. These functions are captured in the system's logical architecture.

The functional framework was then used to develop a set of alternative solutions for a future area-denial system. The individual metrics from the requirements analysis were used to evaluate each of the candidate solutions using a quantitative value model approach [10]. Using these metrics, the AoA scored each of the alternatives on a scale from 0 to 100. The selected solution was the one with the highest score.

The results of the AoA were then used to further build the logical architecture of the system, which captures the behavior and interactions required to perform each system function. The logical architecture was then refined into a physical architecture, which captures the hardware and software components needed to realize those functions. The physical architecture provided a set of technical requirements that were tied back to the initial requirements analysis to ensure traceability.

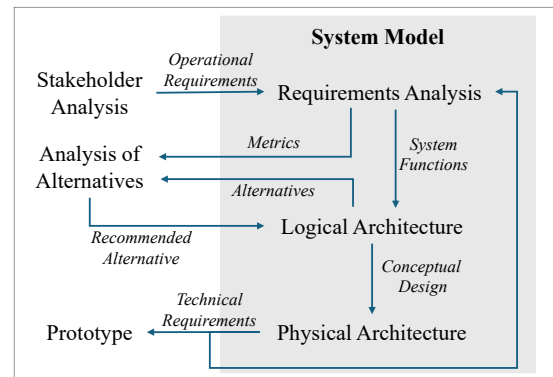


Fig. 2. MBSE-based methodology used for designing a future area denial system.

#### IV. STAKEHOLDER ANALYSIS AND REQUIREMENTS

##### *Stakeholder Analysis*

The stakeholder analysis identified the operational, tactical, and humanitarian needs that shaped the system requirements. Three primary groups participated in this process: combat engineers as operational users, Forward Area operators as mission commanders, and humanitarian organizations as compliance stakeholders. Their priorities were collected through interviews and reviews of representative operational scenarios. The results are summarized in Table I and are discussed below.

Combat engineers serve as the primary deployers of the system, which makes their needs central to the design. They emphasized a modular system that aligns with current doctrine and policy constraints to support mission readiness. Their highest priorities included reliable and accurate target identification supported by documented discrimination criteria that determine whether an intruder is a combatant. They also expressed the need for multi-modal sensing capability, such as acoustic or thermal sensing, to improve confidence in target classification. Finally, they highlighted the importance of a secure and tamper-resistant networking system to reduce the personnel required for monitoring once the system is deployed.

Mission commanders focused on practical considerations related to day-to-day employment. They emphasized that the system must function as an autonomous sentry device capable of monitoring key avenues of approach with minimal training burden. They required lightweight, man-portable components that can be transported and installed quickly by small teams. Cost also emerged as a major concern. Unlike most military equipment, the system is intended to be expendable rather than recovered, which requires the design to minimize procurement cost while maintaining functional reliability.

Humanitarian stakeholders, including international NGOs and legal advisors, centered their concerns on Ottawa Treaty compliance and long-term civilian safety. They required assurance that the system would not create residual hazards once hostilities conclude. Their priorities included safe end-of-life behavior, reliable positive identification to distinguish combatants from civilians and wildlife, and the ability to reverse a deployment without leaving dangerous remnants in the environment. These concerns represent the system's defining distinction from traditional landmines and directly motivate the decision to pursue a non-explosive, sensor-based architecture with a human-in-the-loop function for lethal decisions.

Collectively, the stakeholder analysis informed a set of operational and technical requirements that balance battlefield utility with humanitarian responsibility. This alignment ensures that the proposed system meets combat needs while adhering to international safety and ethical standards.

TABLE I. STAKEHOLDER ANALYSIS RESULTS

| Stakeholder | Key Concerns | Operational Requirement |
|-------------|--------------|-------------------------|
|-------------|--------------|-------------------------|

|                                               |                                                                                                                                          |                                                                                                              |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Combat engineers                              | The sensor will need to operate as a distributed array, resistant to cyber and electronic attack                                         | The system shall maximize resistance to tampering.                                                           |
|                                               | System must be rapidly deployable to the objective.                                                                                      | The system shall minimize deployment time.                                                                   |
|                                               | Recovery of sensors should minimize or all together eliminated                                                                           | The system shall include a method of decomposition to mitigate enemy utilization and non-combatant harm.     |
|                                               | Sensors must remain concealed and non-disruptive.                                                                                        | The system shall minimize radio and electrical signaling as well as maintain a small profile                 |
| Mission commander                             | To protect the soldier, overwatch of the system must be at a safe distance away                                                          | The system shall maximize the distance of usage available for the user to operate.                           |
|                                               | Sensors and all utilized systems must support long mission capabilities; excess of +1 year.                                              | The system shall maximize operational status while on mission                                                |
|                                               | Autonomous capabilities would be needed to limit the number of personnel in the vicinity.                                                | The system shall autonomously operate and deliberate itself before sending a signal out.                     |
| Humanitarian organizations and non-combatants | There cannot be any fallout, or long-term hazards present that can harm local civilians in the vicinity of the deployment of the system. | The system shall not contain hazardous products or biproducts through the use and destruction of said system |
|                                               | All checks cannot be automatic, and some level of human verification must be available before a life is taken                            | The system shall have a human moderator to enable a Human in the loop function.                              |
|                                               | Any autonomous system must discriminate correctly between targets.                                                                       | The system shall have the ability to discriminate between combatants and non-combatants.                     |

##### *B. System Functional Requirements*

The stakeholder analysis directly informed the identification of the high-level functions that the system must perform. These functions were translated into a use-case diagram, which provides a visual representation of how external actors interact with the system. In a use-case diagram, each actor represents a user or external entity that engages with the system, and each use case represents a function or behavior the system must perform to meet stakeholder needs. This diagram therefore links stakeholder requirements to system functionality in a clear and traceable manner.

Figure 3 displays the use-case diagram for the Future Area Denial System. The diagram outlines the primary operational sequence in which the system autonomously senses activity within a designated area, generates a signal once intruder activity is detected, and escalates its response when positive identification is confirmed. The initial system involves sensing activity in the secured zone using sensors emplaced by engineers, while filtering intrusions caused by both hostile forces and non-combatants. Once activity is detected, the system provides the soldier with information on the intruder's composition and intent to support the accurate determination of threat level and response. If the intruder is positively identified and confirmed, the system can escalate to a neutralization action

that remains under explicit human control to meet operational, ethical, and safety constraints.

Each stakeholder contributes to the interactions represented in the diagram. Engineers define how the system is deployed and maintained. Mission commanders act as the operational decision-makers who validate identification and authorize escalation. Civilians and wildlife represent protected entities that the system must detect but never harm. Intruders represent adversarial actors whose presence triggers the system's sensing and decision logic. By capturing these relationships and the associated behaviors, the use-case diagram forms the foundation of the system's logical architecture and ensures that each high-level function can be traced back to an explicit stakeholder need.

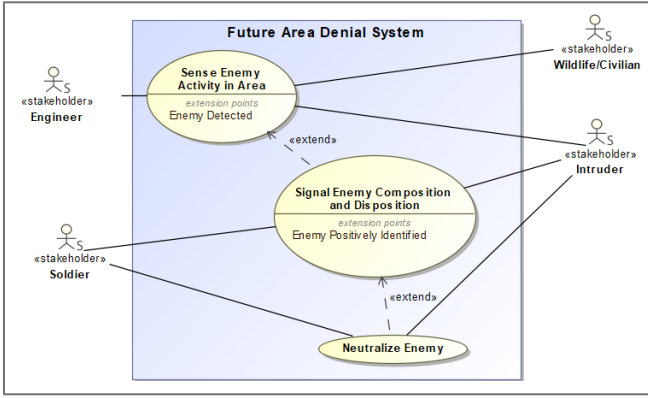


Fig. 3. Use Case diagram for the Future Area Denial System

## V. ANALYSIS OF ALTERNATIVES

### Alternatives

Four alternatives were evaluated to determine the most effective approach for the Future Area Denial System. One existing U.S. military capability, SCATMINES, served as the baseline for comparison. Three additional sensor-based concepts were generated through stakeholder requirements analysis. The development of these novel concepts traces directly from the high-level functions identified in the use-case diagram, which defined the system's core behaviors: sensing intruder activity within a designated area, discriminating between threats and non-threats, and escalating response under human authority when positive identification is confirmed.

The baseline system, SCATMINES, satisfies the area sensing and engagement functions by distributing anti-personnel and anti-tank mines across a target zone. However, it does not provide the discrimination and safe escalation functions required in the use-case diagram. SCATMINES lack long-term safety mechanisms, create humanitarian risk after conflict, and cannot distinguish civilian presence from enemy intrusion. These limitations established the functional gaps that guided the design of the novel alternatives.

The three novel alternatives present distinct operational concepts that map to the functional sequence outlined in the use-case diagram. The Integrated Reconnaissance and Kamikaze Drone System performs continuous aerial sensing of the secured

area and conducts engagement after positive identification. The Silent Triangulation with Call for Fire concept employs distributed inert ground sensors that detect and triangulate intruder movement, then cue indirect fire assets only after sufficient confidence in classification. The Drone Hub with Puck Sensors alternative combines wide-area ground detection with aerial verification to ensure human oversight before any engagement decision.

Although the three novel concepts implement the core system functions in different ways, they share several design priorities derived from stakeholder concerns. All emphasize non-explosive routine operation, positive identification capabilities before engagement, and the ability to deactivate or remove the system without creating post-conflict hazards. Table II summarizes the four alternatives and their associated operational approaches.

TABLE II. ALTERNATIVE SOLUTIONS FOR AREA DENIAL

| Alternatives          | Description                                                                                                                                                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCATMINES             | SCATMINES are scatterable mine delivery systems deployed by artillery, rockets, or aircraft to distribute anti-personnel and anti-tank mines across an area without exposing engineers to enemy fire. They include timed self-destruction and self-deactivation features. |
| Kamikaze Drone System | The Kamikaze system uses a drone to patrol a preset area. When an intrusion is detected, it alerts a drone hub, which launches an explosive-laden drone to engage the target.                                                                                             |
| Silent Triangulation  | A field of inert puck sensors is dispersed by hand or artillery. When they detect an intrusion, they triangulate its location and transmit it to a fires direction center, which coordinates indirect fire on that point.                                                 |
| Drone Hub             | A larger field of inert puck sensors is deployed the same way, but upon detecting an intrusion, a drone is launched from a hub to confirm the target and call for fire through a human operator.                                                                          |

### B. Metrics

Each alternative was evaluated using the metrics listed in Table III. These metrics and their corresponding weights were derived directly from the stakeholder analysis and the operational requirements that followed. Combat engineers prioritized positive identification accuracy, disposability, and compatibility with existing doctrine and equipment. Mission commanders emphasized cost, ease of use, and long-term reliability across repeated deployments. Humanitarian stakeholders required safe end-of-life behavior and compliance with international law. These collective priorities were translated into fourteen quantitative metrics and weighted on a scale of 1 to 4, where 4 represents a critical mission requirement and 1 represents a desirable but secondary characteristic. All relationships among stakeholders, requirements, metrics, and weights were captured within the system model to ensure full traceability.

Each alternative received a score from 0 to 100 for every metric. The 0 to 100 scale for each metric was also derived from stakeholder feedback and operational expectations. A score of 0 reflects a failure to satisfy the stakeholder's needs, and a score of 100 reflects complete satisfaction of the requirement under



realistic conditions. To define the scoring bounds, stakeholders outlined both the minimum acceptable performance and the ideal operational performance for each metric. For example, for positive identification accuracy, a score of 0 corresponds to a system that cannot reliably distinguish combatants from civilians or animals, while a score of 100 corresponds to a system that consistently demonstrates accurate discrimination with near-zero false positives. Scores between these bounds were assigned based on performance relative to stakeholder expectations and technical assessments.

TABLE III. METRIC WEIGHTS AND DESCRIPTIONS

| Metric                          | 0 score                                              | 100 score                                              | Weight |
|---------------------------------|------------------------------------------------------|--------------------------------------------------------|--------|
| PID accuracy                    | Cannot distinguish targets from civilians or animals | Fully distinguishes targets from civilians and animals | 3      |
| Time to Lethal Response         | Fails to meet doctrinal timing                       | Exceeds doctrinal timing by 15 minutes or more         | 1      |
| Accuracy of Lethal Response     | Low targeting accuracy                               | Highly accurate targeting with minimal error           | 2      |
| Resilience to Jamming           | Easily jammed                                        | Very resistant to jamming                              | 2      |
| Price                           | Very expensive to construct                          | Very inexpensive to construct                          | 1      |
| Detection Range                 | Detects only within 5 m                              | Detects at five times that range or more               | 2      |
| Signal Reception Range          | Operator must be within 0.5 miles                    | Operator can be 5 miles away without errors            | 3      |
| Autonomy                        | No autonomy                                          | Fully autonomous                                       | 2      |
| Longevity                       | Works for only one mission                           | Works for five or more missions                        | 2      |
| Reusability                     | Single use                                           | Multiple use                                           | 2      |
| Disposability                   | Cannot be removed or disabled                        | Easily removed or disabled                             | 4      |
| Detectability and Vulnerability | Easily detected and exploited                        | Very hard to detect and exploit                        | 2      |
| Availability of technology      | TRL of 1                                             | TRL of 9                                               | 1      |

### Scoring

Each alternative was evaluated using the 100 point scale for every metric with the scores shown in Table IV. The raw score for a metric was multiplied by its assigned weight to account for the priority of that requirement to the stakeholder community. The weighted values were then normalized so that a total score of 100 represents an ideal system that satisfies all stakeholder and operational expectations. Summing the normalized weighted values yielded the total performance score for each alternative shown in Table IV. This method ensured that the evaluation remained aligned with stakeholder priorities and that every design outcome was traceable to the operational and humanitarian requirements captured in the system model.

From these results, SCATMINES ranked lowest with a total score of 50, largely due to poor performance in positive identification accuracy, signal reception range, and reusability, despite strong marks for price and availability of technology. Silent Triangulation achieved a higher total score of 70, driven by longevity and reusability, but its limited resilience to jamming and slower lethal response time reduced its competitiveness. The two highest scoring alternatives were the

Kamikaze Drone System with a score of 77 and the Drone Hub with Puck Sensors at 79. Both performed well in positive identification accuracy, autonomy, and signal reception range. The Drone Hub with Puck Sensors ranked slightly higher because of superior disposability and longevity, which reflected the strongest humanitarian and operational concerns across the stakeholder community.

Based on these results, along with additional stakeholder discussions and a team review, the Drone Hub with Puck Sensors was selected as the preferred concept for continued development.

TABLE IV. ALTERNATIVE SCORES

| Metric                           | Alternative           |            |                      |           |
|----------------------------------|-----------------------|------------|----------------------|-----------|
|                                  | Kamikaze Drone System | SCAT MINES | Silent Triangulation | Drone Hub |
| PID accuracy                     | 95                    | 10         | 95                   | 95        |
| Time to Lethal Response          | 90                    | 95         | 50                   | 95        |
| Accuracy of Lethal Response      | 95                    | 80         | 80                   | 95        |
| Jammability                      | 30                    | 90         | 20                   | 40        |
| Price                            | 50                    | 85         | 30                   | 40        |
| Detection Range                  | 70                    | 5          | 70                   | 50        |
| Signal Reception Range           | 90                    | 0          | 95                   | 90        |
| Autonomy                         | 99                    | 15         | 60                   | 99        |
| Longevity                        | 80                    | 70         | 90                   | 90        |
| Reusability                      | 99                    | 0          | 90                   | 99        |
| Disposability                    | 85                    | 70         | 85                   | 90        |
| Detectability and Counterability | 40                    | 40         | 40                   | 50        |
| Availability of technology       | 80                    | 100        | 80                   | 80        |
| Ease of integration              | 60                    | 100        | 40                   | 60        |
| <b>Total Weighted Score</b>      | <b>77</b>             | <b>50</b>  | <b>70</b>            | <b>79</b> |

## VI. CONCEPTUAL DESIGN

Following the selection of the Drone Hub with Puck Sensors alternative, the MBSE framework guided the development of detailed system architecture. This section presents the logical architecture describing operational behavior, the physical architecture defining component relationships, and the derived technical requirements that ensure the design satisfies stakeholder needs.

### A. Logical Architecture

The system operates as a continuous, multi-phase autonomous process designed to detect, analyze, and relay environmental data for rapid situational awareness.

As illustrated in Figure 4, the operational cycle begins in the detection phase, where distributed ground-based sensor pucks monitor designated areas using onboard motion sensors and ranging systems. When an anomaly is detected, the sensor puck transitions from standby to alert state, recording geographic coordinates via GPS, and transmitting detection data through its radio to the command node (interface R1).

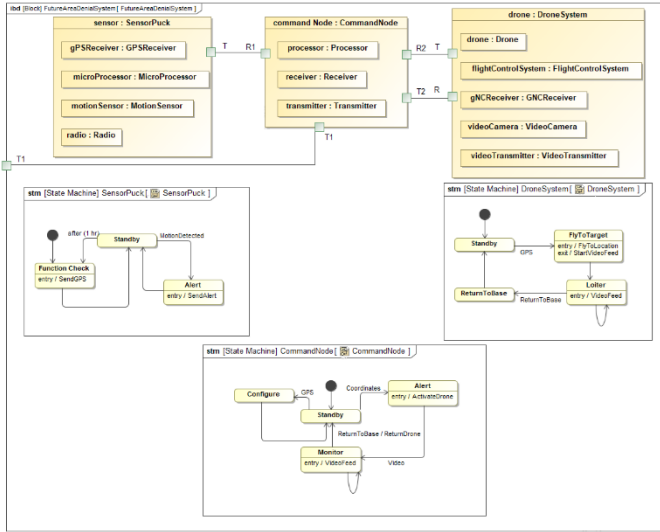


Fig. 4. Logical Conceptual Design.

During the tasking phase, the command node receives the alert, processes the detection coordinates, and assigns the event to an available drone via interface R2. The command node transitions to an alert state, activating the assigned drone through GPS coordinate handoff (interface T2).

The observation phase initiates as the drone autonomously navigates to reported coordinates using GPS-based waypoint guidance. Once on station, the drone activates its onboard camera system and employs AI-based visual detection to identify, classify, and track targets in real time. The drone streams continuous live video and metadata—including GPS coordinates, object classification, and movement data—to Forward Area operators via its video transmitter (interface T1). The command node enters monitor state, relaying the video feed to operators for human-in-the-loop verification.

In the data relay phase, operators monitor live conditions, confirm detections, and provide targeting guidance or redirect the platform for additional coverage. The system maintains this observation posture until the operator confirms mission completion or determines if the detection was a false alarm.

Finally, the recovery phase initiates when the mission concludes. The drone autonomously returns to its hub for data offload, battery recharge, and network reintegration, resetting the entire system to the standby state for the next detection event. Sensor pucks remain in standby mode, continuously monitoring for subsequent intrusions.

This closed-loop architecture ensures persistent area monitoring while maintaining human control in the targeting decision loop, satisfying both operational effectiveness and ethical compliance requirements.

### B. Physical Architecture

The physical architecture defines the hardware and software components required to implement the logical operational flow. The system comprises three primary subsystems: sensor pucks, command node, and drone platforms.

**Sensor Puck Subsystem:** Each sensor puck integrates a GPS receiver for coordinate capture, a motion sensor (mmWave radar or PIR) for intrusion detection, a microprocessor (ESP32) for local processing and state management, and a radio transceiver for mesh networking with the command node. The puck operates on battery power with low-power standby mode to maximize deployment duration.

**Command Node Subsystem:** The command node consists of a central processor (Raspberry Pi) managing detection prioritization and drone tasking, a receiver for ingesting sensor puck alerts via ESP-NOW protocol, and a transmitter for coordinating drone dispatch and video relay. The command node serves as the network hub, managing data flow between distributed sensors and aerial platforms.

**Drone Subsystem:** Each drone integrates a flight control system for autonomous navigation, a GNC (Guidance, Navigation, and Control) receiver for waypoint following, a video camera with onboard AI processing for target classification, and a video transmitter for real-time streaming to operators. Drones operate from a charging hub that supports multiple units for scalability.

Component interfaces (R1, R2, T1, T2, R \*MIGHT CHANGE) enable data exchange between subsystems. Interface R1 transmits detection alerts from sensor pucks to the command node. Interface R2 passes targeting coordinates from the command node to assigned drones. Interface T2 provides GPS waypoint guidance to drone navigation systems. Interface T1 streams live video from drones through the command node to operator displays. Interface R supports bidirectional communication for drone return-to-base commands.

This modular architecture enables independent subsystem development and testing while supporting scalability through the addition of sensor pucks or drone units without redesigning core components.

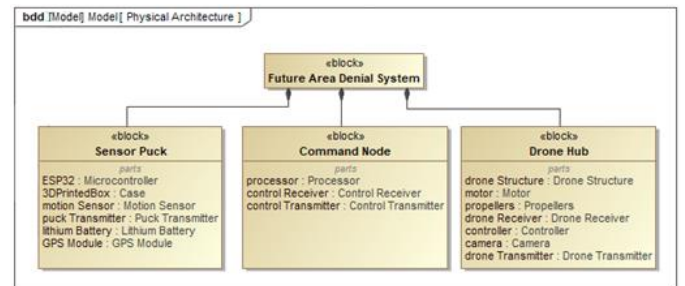


Fig. 5. Block Definition Diagram of the physical architecture of the system

### C. Requirements

Technical requirements were derived systematically from operational requirements identified during stakeholder analysis. These requirements are shown in a requirement diagram on Figure 6, which shows the traceability between the different requirements.

The four foundational requirements driving the system design are:

REQ-1: Autonomous Navigation and Identification - The system shall autonomously detect, navigate to, and classify intrusions without operator intervention during detection and transit phases. Derived from combat engineer need for reduced manpower and common soldier need for sentry functionality.

REQ-2: Operator Reporting - The system shall provide real-time video feed and metadata (GPS coordinates, classification, timestamp) to Forward Area operators for verification and targeting decisions. Derived from humanitarian stakeholder need for positive identification and combat engineer need for human-in-the-loop verification.

REQ-3: Self-Expiration - The system shall include programmable deactivation timers or remote disable capability to ensure safe end-of-life behavior. Derived from humanitarian stakeholder requirement for Ottawa Treaty compliance and post-conflict safety.

REQ-4: Low Signature - The system shall maintain minimal visual, acoustic, and electromagnetic signatures to avoid detection and counteraction by adversaries. Derived from combat engineer need for tamper resistance and operational requirement for survivability.

Supporting requirements address deployment speed (REQ-5: 24-hour emplacement), environmental tolerance (REQ-6: operation in -20°C to 50°C), communication range (REQ-7: 5km line-of-sight), power endurance (REQ-8: 72-hour minimum sensor puck operation), modularity (REQ-9: field-swappable components), cost constraint (REQ-10: <\$500 per sensor puck), ease of use (REQ-11: <2 hours training), and non-recovery design (REQ-12: expendable components with data sanitization).

Most requirements will be verified through component-level testing (hardware bench tests, software unit tests) and validated through integrated system demonstrations. However, several requirements fall outside the project scope for full validation. REQ-3 (self-expiration) and REQ-5 (24-hour deployment) require extended field testing in operational environments. REQ-12 (non-recovery) raises security concerns if adversaries capture components; future work must address data encryption and tamper-evident designs to mitigate this risk while maintaining the expendable design philosophy.

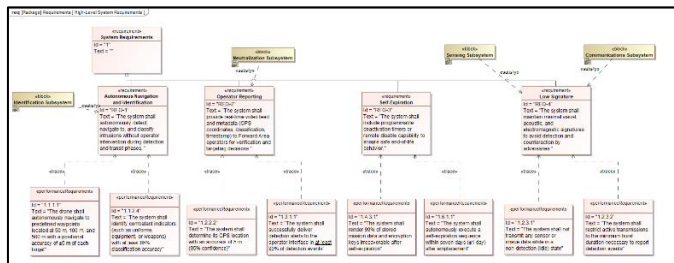


Fig. 6. Requirement diagram

## VII. CONCLUSION

This research demonstrates the application of MBSE for the design of a safe, sensor-based alternative to traditional minefields. Through stakeholder analysis, requirements analysis, an AoA, and system architecting the project identifies a feasible and ethically compliant solution that integrates

ground sensors with autonomous aerial overwatch. This hybrid configuration provides scalable, precise, and reversible area denial capabilities while eliminating many humanitarian and legal risks associated with explosive ordnance systems.

The selected Drone Hub with Puck Sensors concept proved to be the optimal balance of effectiveness, cost, and sustainability. It enables persistent monitoring, rapid detection, and human-in-the-loop verification while maintaining compliance with the Ottawa Treaty and other international standards. The system's modular architecture allows for future improvements in sensing, communications, and autonomy as technology advances.

Future work will focus on expanding prototype testing to validate system performance in realistic field conditions, regarding both deployment and coordination of the system. There will also be a focus on exploring enhanced power management and communication resilience to improve overall system performance capabilities. This project provides a scalable framework for future non-lethal area denial systems that align with modern operational needs and ethical imperatives of modern warfare.

## ACKNOWLEDGMENT

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